

Autonomous Mobile Robot

SLAM Mapping & Mission Execution in a City-Town Simulation

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Simulation Environment

World

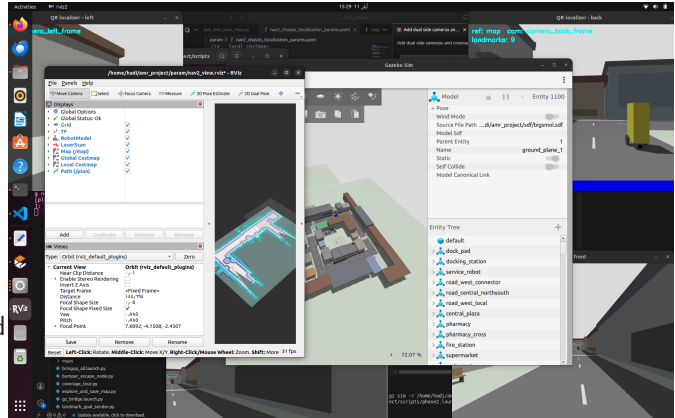
Urban city-town map with 9 QR-tagged landmarks

Robot

- Diff-drive chassis, $2.0 \times 1.0 \times 0.5$ m
- 4 cameras, 360° planar lidar, IMU, 4 bumpers

Dynamic obstacles

- Two scripted patrol bots looping fixed waypoints



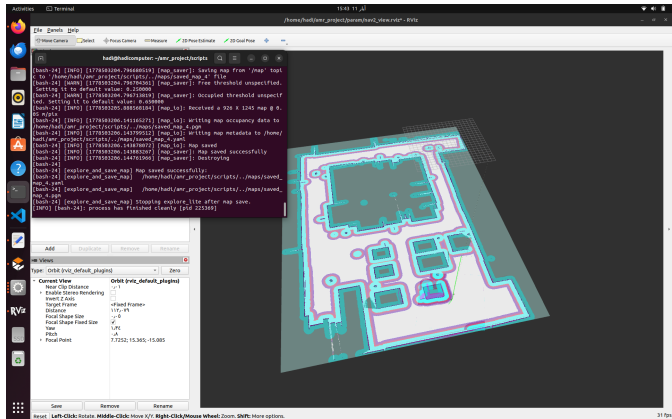
SLAM: Mapping the Environment

Mapping pipeline

- `slam_toolbox (async)` builds the map from `/scan`
- `explore_lite` drives autonomously to frontiers
- EKF fuses wheel odometry + IMU

Landmark detection

- QR detection on all 4 cameras
- Bearing (camera) + range (lidar) fusion
- Saved to landmarks.yaml



Mission Control Interface

Tkinter GUI

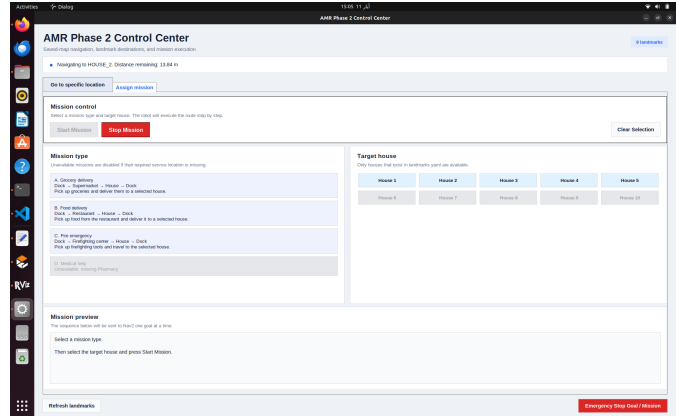
- Direct navigation to any landmark
- Pre-defined multi-step missions

Missions

- Grocery, Food, Fire Emergency
- Each is DOCK → stop → HOUSE → DOCK

Robustness

- Watchdog cancels & retries if stuck >30 s
- Multi-leg missions continue past a failed leg



Dynamic Obstacle Avoidance

Costmap layers

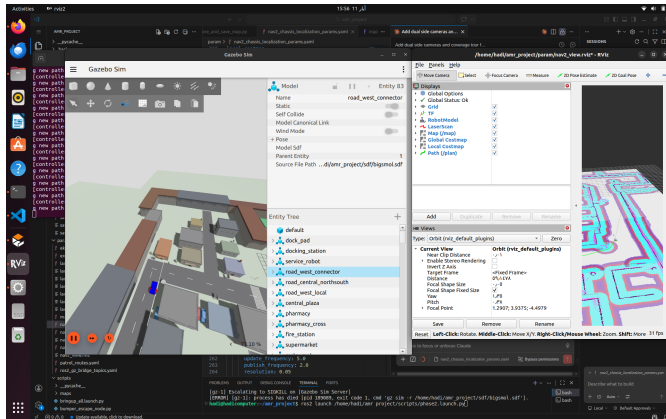
- Static walls + live /scan obstacles
- Dual inflation: walls 1.5 m, dynamics 0.3 m

Controller: MPPI

- 1200 trajectories / cycle, 2 s horizon
- Auto-disables PathAlign when path is blocked

Replanning

- BT re-plans at ~ 1 Hz
- Collision monitor as safety fallback



Thank you

Questions?



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